

$$h) \quad h.1) \quad a. \quad \cos^2 u = \frac{\cos(2u) + 1}{2}$$

$$(a) \quad 2\cos^2(u) = \cos(2u) + 1$$

$$(b) \quad 2\cos^2(u) = \cos^2(u) - \sin^2(u) + 1$$

$$(c) \quad \cos^2(u) = 1 - \sin^2(u)$$

$$(d) \quad \cos^2(u) = \cos^2(u)$$

$$\cos(2u) = \cos^2(u) - \sin^2(u)$$

$$\sin^2 + \cos^2(u) = 1$$

$$\therefore \cos^2(u) = 1 - \sin^2(u)$$

$$C- \quad \underline{\cos(3u) = 4\cos^3(u) - 3\cos(u)}$$

$$\cos(3u) = \cos(2u + u)$$

$$\cos(2u) = \cos^2(u) - \sin^2(u)$$

$$\sin(2u) = 2\sin(u)\cos(u)$$

$$\underline{\quad \quad \quad} \quad \underline{\quad \quad \quad}$$

$$\cos(2u + u) = \cos(2u)\cos(u) - \sin(2u)\sin(u)$$

logo

$$\cos(3u) = (\cos^2 u - \sin^2(u))\cos(u)$$

$$= - (2\sin(u)\cos(u))\sin(u)$$

$$= \cos^3 u - \sin^2 \cos(u) - 2\sin^2(u)\cos(u)$$

$$= \cos^3(u) - 3\sin^2(u)\cos(u)$$

$$\begin{aligned}
 &= \cos^3(u) - 3(1 - \cos^2 u)(\cos u) \\
 &= \cos^3(u) - 3(\cos u - \cos^3(u)) \\
 &= 4\cos^3(u) - 3\cos(u)
 \end{aligned}$$

C. g. p.

4.2) a- $\arcsin\left(\sin \frac{5\pi}{4}\right)$

$\arcsin x = y \Leftrightarrow \sin y = x, x \in [-\pi/2, \pi/2]$

Qual é
o ângulo
cujo seno
é igual a y

$$\begin{aligned}
 \sin \frac{5\pi}{4} &= \sin\left(\pi + \frac{\pi}{4}\right) = -\sin\left(\frac{\pi}{4}\right) \\
 &= -\frac{\sqrt{2}}{2}
 \end{aligned}$$

$$\arcsin\left(\sin\left(\frac{5\pi}{4}\right)\right) = \arcsin\left(-\frac{\sqrt{2}}{2}\right)$$

Sabemos que $\sin\left(-\frac{\pi}{4}\right) = -\frac{\sqrt{2}}{2}$

logo $\arcsin\left(-\frac{\sqrt{2}}{2}\right) = -\frac{\pi}{4}$

$$-\frac{\pi}{4} \in [-\pi/2, \pi/2]$$

d- $\arcsin\left(\sin\left(-\frac{\pi}{6}\right)\right) = -\frac{\pi}{6} \in [-\pi/2, \pi/2]$

logo $\arcsin\left(\sin\left(-\frac{\pi}{6}\right)\right) = -\pi/6$

$$e - \arcsin(\sin \frac{23\pi}{6})$$

$$\begin{aligned} \sin\left(\frac{23\pi}{6}\right) &= \sin\left(\frac{24\pi}{6} - \frac{\pi}{6}\right) = \sin\left(4\pi - \frac{\pi}{6}\right) \\ &= \sin\left(-\frac{\pi}{6}\right) \end{aligned}$$

$$\arcsin(\sin -\pi/6)$$

$$= -\frac{\pi}{6}, \text{ pois } -\pi/6 \in [-\pi/2, \pi/2]$$

$$k - \operatorname{tg}(\operatorname{arccotg} 3)$$

$$\theta = \operatorname{arccotg} 3, \operatorname{cotg} \theta = 3 \text{ e } \theta \in]0, \pi[$$

$$\operatorname{tg} \theta = \frac{1}{\operatorname{cotg} \theta} = \frac{1}{3}$$

$$\text{Logo } \tan(\operatorname{arccotg} 3) = \frac{1}{3}$$

$$l - \arctg(\operatorname{cotg} \frac{\pi}{5})$$

$$\sin(\pi/2 - a) = \cos a \text{ e } \cos(\pi/2 - a) = \sin a$$

$$\operatorname{tg}(\pi/2 - \theta) = \frac{\cos \theta}{\sin(\theta)} = \operatorname{cotg} \theta$$

$$\begin{aligned} \text{Assim, } \arctg(\operatorname{cotg} \frac{\pi}{5}) &= \arctg(\operatorname{tg}(\frac{\pi}{2} - \frac{\pi}{5})) \\ &= \arctg(\operatorname{tg}(\frac{3\pi}{10})) \end{aligned}$$

$$\text{Como } \frac{3\pi}{10} \in]-\pi/2, \pi/2[\quad \arctg(\operatorname{tg}(\frac{3\pi}{10})) = \frac{3\pi}{10}$$

$$4.4) \quad a - \arcsin\left(-\frac{\sqrt{2}}{2}\right) = -\frac{\pi}{4} \quad | \quad -\frac{\pi}{4} \in [-\pi/2, \pi/2]$$

$$b - \cotg(\arcsin(-\frac{4}{5}))$$

$$\text{Seja } \theta = \arcsin\left(-\frac{4}{5}\right)$$

$$\Leftrightarrow \sin \theta = -\frac{4}{5} \quad \text{e} \quad \theta \in [-\pi/2, \pi/2]$$

$$\cos \theta = \sqrt{1 - \sin^2 \theta}, \quad \cos \geq 0$$

$$= \sqrt{1 - \frac{16}{25}} = \frac{3}{5}$$

$$\cotg \theta = \frac{\cos \theta}{\sin \theta} = \frac{3/5}{-4/5} = -\frac{3}{4}$$

$$c - \cos(\arcsin 1/2 - \arccos 3/5)$$

$$\text{Seja } A = \arcsin 1/2 \Rightarrow \sin A = 1/2$$

$$\text{e } \cos A = \frac{\sqrt{3}}{2}$$

$$\text{Logo } A = \pi/6$$

$$\text{Seja } B = \arccos \frac{3}{5} \Rightarrow \cos B = \frac{3}{5} \quad \text{e } \sin B = \frac{4}{5}$$

$$\cos^2 \alpha + \sin^2 \alpha = 1$$

$$\sin^2 \alpha = 1 - \frac{9}{25} = \frac{16}{25}$$

$$\sin \alpha = \frac{4}{5}$$

$$\begin{aligned}\text{Usando } \cos(A-B) &= \cos A \cos B + \sin A \sin B \\ &= \frac{\sqrt{3}}{2} \cdot \frac{3}{5} + \frac{1}{2} \cdot \frac{4}{5} = \frac{3\sqrt{3}+4}{10}\end{aligned}$$

$$4.4) \text{ e- } \cos^2\left(\frac{1}{2} \arccos \frac{1}{3}\right) - \sin^2\left(\frac{1}{2} \arccos \frac{1}{3}\right)$$

$$\text{Seja } \theta = \arccos\left(\frac{1}{3}\right)$$

$$\cos^2\left(\frac{\theta}{2}\right) - \sin^2\left(\frac{\theta}{2}\right)$$

$$\cos^2 \alpha - \sin^2 \alpha = \cos(2\alpha)$$

$$\begin{aligned}\text{Com } \alpha &= \frac{\theta}{2} \text{ obtemos } \cos^2\left(\frac{\theta}{2}\right) - \sin^2\left(\frac{\theta}{2}\right) \\ &= \cos(\theta)\end{aligned}$$

$$\text{Como } \theta = \arccos(1/3), \text{ então}$$

$$\cos \theta = \frac{1}{3}$$

$$\begin{aligned}\text{ou seja } \cos^2\left[\frac{1}{2} \arccos\left(\frac{1}{3}\right)\right] - \sin^2\left[\frac{1}{2} \arccos\left(\frac{1}{3}\right)\right] \\ = \frac{1}{3}\end{aligned}$$

- 4.6) • Contínua em $]-\infty, 1]$: $f(u) = 0$ e' constante, logo e' contínua
- Em $]-1, 1[$: $\arccos u$ e' contínua
 - Em $]1, +\infty[$: $\pi/2 \sin(\pi/2 \cdot u)$ e' contínua

$$\text{Em } u = -1, \quad f(-1) = 0$$

$$\lim_{u \rightarrow -1^+} f(u) = \arcsen(1) = -\frac{\pi}{2}$$

$$\lim_{u \rightarrow -1^-} f(u) = 0$$

Logo f é contínua em $u = -1$

$$\text{Em } u = 1$$

$$\lim_{u \rightarrow 1^+} f(u) = \frac{\pi}{2} \arcsen\left(\frac{\pi}{2} \cdot 1\right) = \pi/2$$

$$\lim_{u \rightarrow 1^-} f(u) = \arcsen(1) = \pi/2$$

$$f(1) = \frac{\pi}{2} \arcsen\left(\frac{\pi}{2} \cdot 1\right) = \pi/2$$

f é contínua em $u = 1$

f é contínua em $\mathbb{R} \setminus \{-1, 1\}$

$$u - \text{Dom } f = [-\pi/2, \pi/2]$$

$$\cdot \text{ No ramo } -1 < u < 1, \quad f(u) = \arcsen u \text{ é }]-\pi/2, \pi/2[$$

$$\cdot \text{ No ramo } u > 1, \quad f(u) = -\frac{\pi}{2} \arcsen\left(\frac{\pi}{2} u\right) \in]-\pi/2, \pi/2]$$

$$\cdot \text{ No ramo } u \leq -1, \quad f(u) = 0$$

$$c - \lim_{u \rightarrow -\infty} f(u) = 0, \quad \lim_{u \rightarrow +\infty} f(u) = \infty \text{ exists}$$

$$h. 8) a - e^u = e^{1-u}$$

i. implies

log both sides

$$u = 1 - u$$

$$(A) u = \frac{1}{2}$$

$$d - e^{2u} + 2e^u - 3 = 0$$

$$\text{Set } y = e^u$$

$$y^2 + 2y - 3 = 0$$

$$y = -3 \quad \text{impossible (} e^u > 0 \text{)}$$

$$y = 1 \quad e^u = 1 \quad \text{or } u = 0$$

$$4.9) b - \cosh y + \sinh x = e^u$$

$$\cosh u + \sinh u$$

$$= \frac{e^u + e^{-u} + e^u - e^{-u}}{2} = e^u$$

$$e - \sinh(x+y) = \sinh x + \cosh y + \cosh x + \sinh y$$

$$\begin{aligned}
 & \sinh x \cosh y + \cosh x \sinh y \\
 &= \left(\frac{e^x - e^{-x}}{2} \right) \left(\frac{e^y + e^{-y}}{2} \right) \\
 & \quad + \left(\frac{e^y + e^{-y}}{2} \right) \left(\frac{e^x - e^{-x}}{2} \right) \\
 &= \frac{1}{2} (e^{x+y} - e^{-(x+y)})
 \end{aligned}$$

Pela definição

$$\sinh(x+y) = \frac{1}{2} (e^{x+y} - e^{-(x+y)})$$

4.10) a - argh $x = \ln(x + \sqrt{x^2 + 1})$, $x \in \mathbb{R}$

Seja $y = \operatorname{arcsinh} x$

Por definição $x = \sinh y = \frac{e^y - e^{-y}}{2}$

Então

$$\begin{aligned}
 2x &= e^y - e^{-y} \Rightarrow 2x e^y = e^{2y} - 1 \\
 \Rightarrow e^{2y} - 2x e^y - 1 &= 0
 \end{aligned}$$

$$u = e^y > 0 \quad u^2 - 2xu - 1 = 0$$

As soluções são:

$$u = \frac{2x \pm \sqrt{(-2x)^2 + 4}}{2}$$

$$u = x \pm \sqrt{x^2 + 1}$$

$$\text{Como } u = e^y > 0$$

$$u = x + \sqrt{x^2 + 1}$$

$$e^y = x + \sqrt{x^2 + 1}$$

$$\text{Logo } y = \ln(x + \sqrt{x^2 + 1})$$

$$y = \operatorname{arcsinh} x = \ln(x + \sqrt{x^2 + 1}),$$

